

An Artifact for the Article:

“Generating Realistic Vulnerabilities via Neural Code Editing: An Empirical Study”

ABSTRACT

This artifact contains functional versions of the neural code editors, vulnerability detectors, and their respective datasets for the empirical study. As our empirical study involves 5 different tools (3 neural code editors and 2 vulnerability detectors) which make the setting up complex, we have prepared a Docker image on Zenodo [1] that has already contained necessary components to execute the tools and prompts to help users compare the respective experiment results in the paper. However, we still provide an independent version of the artifact on Zenodo for the users who do not want to use Docker. As our empirical study involves a large amount of data, it is recommended that the host machine has at least 200GB hard disk space, 32GB memory, and NVIDIA GPUs that CUDA 11.1 supports.

Our empirical experiments presented in the paper can be divided into 2 different parts. The first part is an evaluation of the state-of-the-art neural code editors for generating software vulnerabilities which is presented in the results for RQ1 and RQ2 as found in our research paper [2]. In this part, three state-of-the-art neural code editors (Graph2Edit, SequenceR, and Hoppity) are evaluated on a synthetic dataset SARD, two datasets modified from SARD, and one real-world dataset. The second part evaluates whether the generated realistic vulnerabilities in RQ2 can indeed improve the chosen deep learning-based vulnerability detectors, which is presented in the results for RQ3 as found in our research paper. In this part, two state-of-the-art vulnerability detectors, Devign, and ReVeal, are used to test whether the vulnerability datasets with the added generated realistic vulnerabilities can train the detectors better.

As all the 5 evaluated tools are deep learning-based, their models take considerable amount of time (e.g., several days) and hardware resources (e.g., ≥ 32 GB CPU memory and ≥ 24 GB GPU memory) to train. Thus, to make it easier for users to reproduce the experiments, we have saved the trained models for all the tools and the users can test the models on the testing datasets directly. However, we still provide the code, scripts as well as the datasets for training the models from scratch.

Besides, our study also involves manual reviews and user studies. The raw data and results are also provided in our artifact.

[1] <https://zenodo.org/record/6897604>

[2] https://www.researchgate.net/profile/Haipeng-Cai/publication/361835991_Generating_Realistic_Vulnerabilities_via_Neural_Code_Editing_An_Empirical_Study/links/62c7c14000d0b451103f1483/Generating-Realistic-Vulnerabilities-via-Neural-Code-Editing-An-Empirical-Study.pdf